NexTimber® by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

Product Site Guide

Version 1.0
July 2024

Contents

Purpose of Document

Timberlink is the supplier of NeXTimber® Cross Laminated Timber (CLT) and Glue Laminated Timber (GLT) products and associated accessories. Responsibility and governance for the handling, storage, unloading and installation of supplied materials sits with the customer. The information in this guide is specific to NeXTimber® products and is not transferrable to other CLT or GLT suppliers.

Timber is a natural product and susceptible to moisture and exposure to the elements.

This document, including but not limited to as it relates to the safe handling, unloading, care and maintenance of NeXTimber® products, is not an instruction, is non-exhaustive and is intended to be informative only.

This document attempts to address some common challenges anticipated to be faced during construction and provide suggestions for consideration by the customer which may reduce the likelihood of substantial remediation, however the care, installation and maintenance of the product remains the sole responsibility of the customer once the product has been delivered by NeXTimber® .

As per AS 1720.1, "The structure and its structural elements (including timber, metal, adhesives and any other structural material) shall be designed. Any assumed maintenance program shall be specified in order to satisfy strength, stability and serviceability requirements, for the design life of the structure.

Due consideration shall be given to environmental conditions, such as the thermal, physical, chemical, mechanical, and biological agents that may act on the structure to reduce its performance characteristics", making designers and users of NeXTimber® products responsible for how these products are designed, specified, used and maintained.

NeXTimber® cannot assure that seasoning checks will not occur as the result of fluctuations in moisture content at the manufacturing or intermediate site storage and variations in climatic conditions over the course of the construction phase, however, understanding some of the items outlined in this document may influence some decisions made by the designers during the design phase and assist in reducing the severity of discoloration and adverse weather conditions.

Unless otherwise defined in this document, defined terms have the same meaning as in the NeXTimber® Terms & Conditions of Supply.

Safety

This document provides general information only relating to the delivery process, unloading, manual handling and installation of NeXTimber® products and is not a substitute for professional advice. NeXTimber® takes no responsibility for site safety, unloading, handling, installation and maintenance of NeXTimber® products.

NeXTimber® products are heavy and require a crane and/or the use of heavy plant to unload, handle and install. The customer should use licensed riggers, well maintained and certified rigging and suitably qualified and experienced professionals for the safe handling, erection, propping and installation of NeXTimber® products.

The customer is responsible for compliance with all site related work, health and safety requirements and compliance with all relevant legislation. This includes but is not limited to unloading from delivery trucks, installation/assembly, temporary propping, lifting and handling of NeXTimber® products.

To minimise the risk of fire or damage to the timber products, hot works (ie. welding, soldering, grinding and sparks) should be avoided where possible and if absolutely necessary, adequate protection and insulation to the timber surfaces should be provided.

Erection and Installation Responsibility

Prior to the delivery of product by NeXTimber® , the customer should, at a minimum, engage its own temporary works engineer to advise on and approve the safe lifting, installation, and assembly of NeXTimber® products. At a minimum, the temporary works engineer should advise on temporary lifting points and temporary propping required to safely install NeXTimber® products.

Temporary lifting points should be placed with adequate edge distance and spaced appropriately around the centre of gravity and be positioned in such a way as to avoid excessive deformations and stresses. Temporary propping should be specified to ensure that the structure remains stable during the various stages of assembly.

The installation and removal of temporary propping and lifting requirements are entirely the responsibility of the customer and NeXTimber® will not advise on these.

NeXTimber® Project Support

Project Coordinator

Each NeXTimber® project is assigned a Project Coordinator to coordinate the project on behalf of NeXTimber®. This includes handling customer communications, sharing project documents, overseeing manufacturing and coordinating the Delivery Schedule.

Any changes to the project schedule or plans must be communicated to the Project Coordinator promptly.

Should changes be required, the project may incur additional costs which may affect the delivery program. Payment and reconciliation of invoices through the project delivery phase will be managed through the Project Coordinator.

Transportation

NeXTimber® products are typically transported on flatbed trucks. Permits and approvals for the transportation can be arranged by NeXTimber® as part of our logistics contract. Site traffic management and obtaining other permits is the responsibility of the customer.

Prior to commencing the digital manufacturing modelling phase, the customer shall determine and communicate to NeXTimber® any site constraints and vehicle restrictions that may affect the logistics plan, including without limitation the size of trucks that can be accepted per day for delivery.

Factory Loading

NeXTimber® panels will be supplied strapped together in bundles with multiple elements stacked on top of one another separated by timber shims. Generally, large panels will be loaded individually.

Bundles of panels and large individual panels will (unless otherwise requested by the customer and agreed to by NeXTimber®) be delivered factory-wrapped. The wrapping is for temporary protection only during transport. If materials are to be stored either on site or otherwise the customer will need to arrange additional protection, *refer to section 'Temporary Storage'.*

Bundles or large panels are usually separated by dunnage on the truck for unloading by the customer.

Bundles are labelled with the individual panels, the panel weights and the total weight of the bundle. The customer must advise NeXTimber® of the maximum bundle dimensions (size and weight) for their preferred unloading methodology prior to NeXTimber® commencing the manufacturing modelling phase.

Where practical, the panels will be loaded according to the proposed Delivery Sequence. This is to be determined with the customer prior to commencing digital manufacturing modelling. Some sorting and staging on-site may still be required as the final load order may not follow the Delivery Sequence due to safety requirements or other reasonable concerns that may arise during loading at the factory.

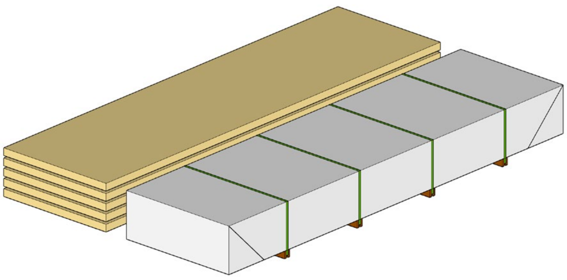

Figure 1: Example of stacked and wrapped panels

Delivery Arrival and Acceptance

Delivery Documentation

The following documentation is supplied with each delivery:

- Panel arrangement plans;
- Truck loading documentation;
- Packaging and pack weights; and
- Panel numbers and sequences.

Panels are labelled with their own unique code and identifier. These correlate to the panel arrangement plans, truck loading documents and NeXTimber® modelling documentation.

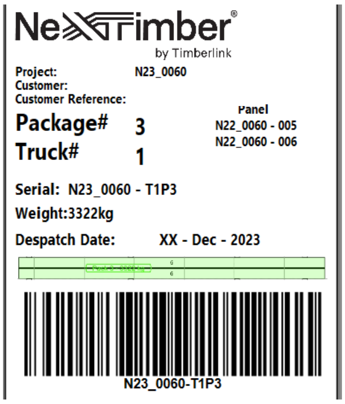

Figure 2: Example of labelling on pack

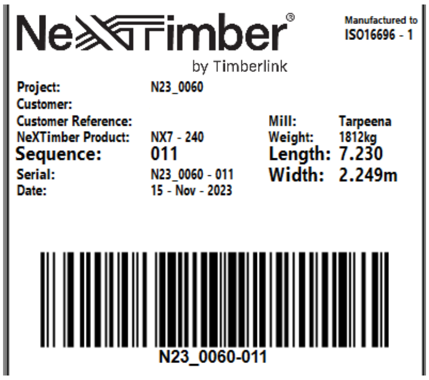

Figure 3: Example of labelling on panel

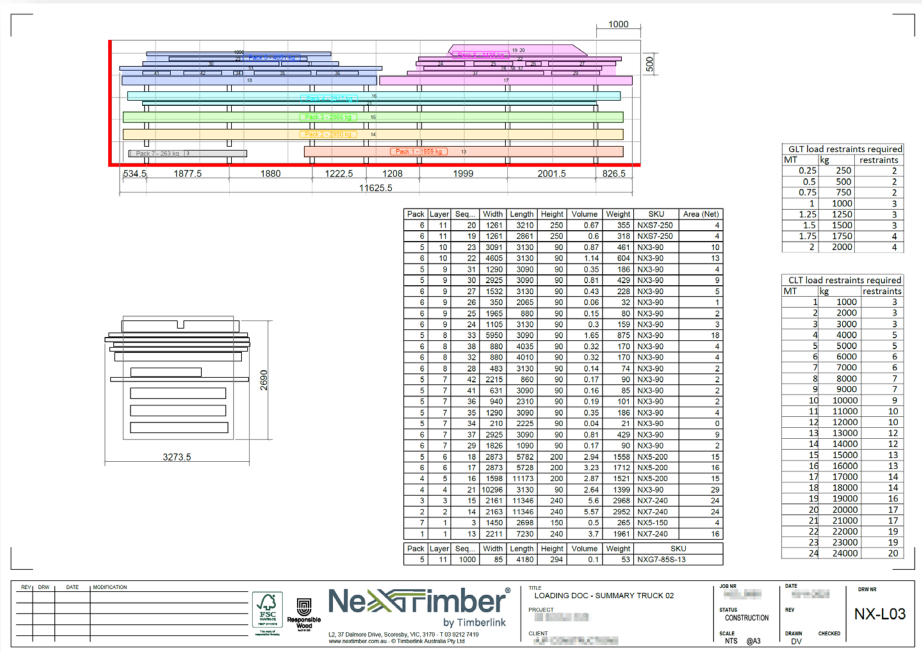

Figure 4: Example of Loading / Sequence Drawings

Receipt of Goods

The customer is responsible for providing clear, easily accessible and unimpeded site access and a suitable staging area to unload the truck, as well as an area for the driver(s) of delivery vehicle(s) to remain safe during the customer's unloading process.

Before unloading from the truck, the customer is to inspect the products for any visible defects or damage (such as torn or damaged wrapping, dents, abrasions etc) and check them against the Despatch Docket for any discrepancy between the products delivered (including quantities) and products stated in the Despatch Docket.

If any issues are identified, the customer shall take a photographic record of the issue or damage and notify NeXTimber® immediately but no later than one business day of delivery.

Temporary Storage / Delays

NeXTimber® products ideally are manufactured on a just-in-time basis, such that delivery is to occur immediately following manufacture. Should there be site delays, it may be possible to delay delivery with sufficient notice prior to NeXTimber® delivery vehicle(s) leaving for the site and will need to be organised with the NeXTimber® Project Coordinator.

Please be aware however that in such scenarios, storage of products will be at the customer's risk and cost and the customer will subsequently be deemed to have accepted the products immediately upon delivery.

If factory despatch cannot be delayed by NeXTimber® , accommodations for offsite storage may be required to be made by the customer. When considering offsite storage, we recommend it is within an indoor environment or under a weatherproof roof structure, elevated off the ground and protected from moisture and direct sunlight.

The protection and handling of NeXTimber® materials in the offsite storage facility is the responsibility of the customer and the customer will need to ensure that the facility has sufficient plant and equipment to accept the goods.

Unloading NeXTimber® Products

The products need to be unloaded by the customer within 2 hours of arrival (unless previously agreed), *refer section 'Delivery Arrival and Acceptance'*. Should the customer take longer than this, it will be liable to NeXTimber® for any resulting extra costs incurred by NeXTimber® .

Prior to commencing unloading, the customer is to ensure that:

- any forklift being used has long enough tines, adequate capacity and maneuverability;
- any crane has adequate capacity and suitably approved rigging;
- all lifting devices, including those attached to any products, are appropriate for their intended use and securely fitted; and
- any person engaged by the customer in connection with unloading the products holds all necessary licences, qualifications and certifications.

If unloading with a forklift, it is likely that extended tines will be required to handle the width of the panels. If this method is going to be used, NeXTimber® needs to be notified prior to the digital manufacturing modelling process to ensure panel dimensions and pack sizes are suitable.

Handling & Protection

- Unload trucks and handle panels using only well-maintained and fit-for-purpose lifting equipment. Never drag, dump, or drop panels.
- Use wide fabric slings that will not mark or damage the timber. The use of chains or steel cables is not recommended and should be avoided.
- When unloading or moving panels, direct contact between the panel and fork surface should be avoided. Special care is to be taken when dealing with visual panels, any marks or damage following delivery, including during unloading, is the risk and responsibility of the customer.
- If the client determines that use of chains or steel cables is absolutely necessary, the customer must ensure protective blocking, padding corner and edge protection to minimise accidental damage is utilised to protect panels.
- Walking on unprotected panels and/or handling the panels with dirty hands or equipment should be avoided. This has the potential to mark or stain the timber.

Tips for Caring for NeXTimber® CLT and GLT

Unless requested by the customer and agreed to by NeXTimber®, NeXTimber® products come wrapped from our factory for temporary protection during transportation.

This wrapping is a temporary cover only and is not guaranteed to be watertight, so where panels need to be temporarily stored, they should be covered by the customer with a covering to protect from rain, moisture and UV.

- Do not place the products directly onto the ground or slab or allow them to have direct contact with moist or wet surfaces as doing so may lead to the uptake of moisture and propagation of the development of mould and mildew.
- Stack product on a dry, free draining and stable staging surface, ideally elevated on 300mm high timber bearers equal or greater than the width of the packs to allow sufficient air flow around the product. Do not allow water to pool under the product where it can evaporate and accumulate on surface of the product.
- Panels should be supported at maximum 3m centres and blocking between stacked panels is to align for the full height of the stack to create a direct load path to the ground.
- Upon arrival if the product is wrapped on the underside, the bottom of the packs should be slit or punctured to ensure that any accumulated moisture can escape the packaging.

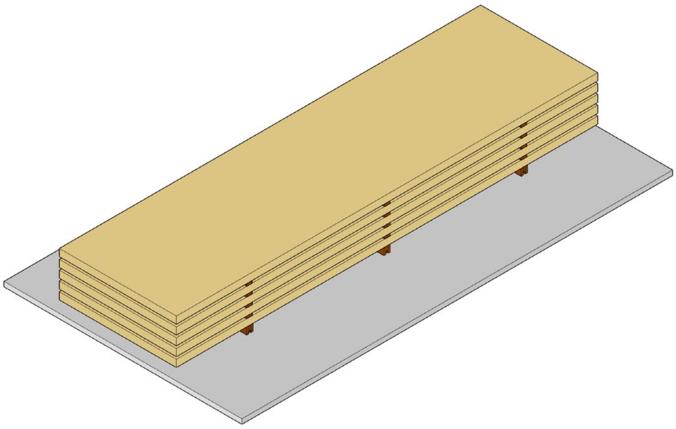

Figure 5: Image of vertically stacked product, separated by packing strips and bearers.

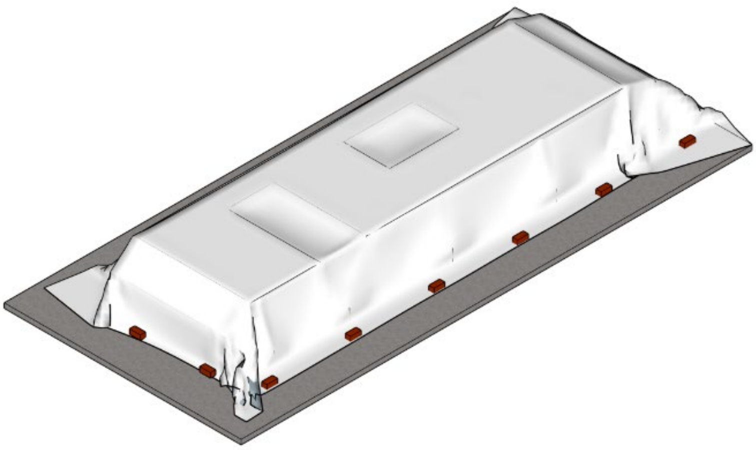

Figure 6: Image of goods, protected with loose fitting, water protective temporary cover.

Weathering

NeXTimber® products are manufactured from natural timber and by their nature, are susceptible to damage if exposed to adverse weather conditions.

The following is a list of some of the suggested precautions that may be taken to protect the timber during the construction phase. It is also worth considering the acceptable level of superficial marks from fingerprints (oil and grease stains from gloves etc.), shoe marks and manual handling damage and the extent of protection required to minimise any remedial actions should it be required.

Weather and Site Water Management

Untreated timber products can withstand temporary exposure to the elements during the construction phase however good construction practices include the implementation of a risk assessment and proactive weather and moisture management plan.

Taking appropriate steps to manage storm water may reduce the possibility and severity of cracking and checking, however NeXTimber® cannot assure that seasoning checks will not occur as the result of fluctuations in moisture content at the manufacturing or intermediate site storage and variations in climatic conditions over the course of the construction phase.

It is also beneficial to prepare a moisture management plan as part of the customer's design and documentation process, so that should the management plan deem that temporary drainage points or similar are necessary, there is sufficient time that these can be incorporated into the timber manufacturing documentation process.

Potential Impacts of Weathering

Some common issues that arise from weathering may include:

- Changes in colour due to UV exposure,
- Water staining and moisture wicking (this can be difficult to remove),
- Saturated surfaces may experience cupping and distortion that will require sanding,
- Development of mould and mildew,
- Checking, splits and cracks,
- Stresses on connections as a result of excessive shrinking and swelling, and
- Dimensional issues.

Some things to consider when developing the project's weather management plan are:

- Maintenance of overall site safety,
- The cost and time implications of remediation works required to rejuvenate timber elements,
- What level of weather effected timber elements are acceptable,
- Whether all exposed timber components require touch-ups prior to applying surface finishes,
- Structural performance due to weather and moisture exposure – particularly around joints and connections, and
- Cost and time allowance for rejuvenation work.

Preventative Strategies to Consider

There is no one-size-fits-all approach for managing site moisture (eg. stormwater) and the extent of strategies adopted will generally be a balance between cost and risk and whether the product will remain exposed during its service life.

However, projects that have adopted a proactive approach to reducing the impacts of rain and other moisture sources have demonstrated that they reduce the overall construction program, accelerate follow-on trades and reduce the extent of rejuvenation works. Some examples may include:

- Co-ordinating the delivery and installation sequence to minimise the requirement for site or offsite storage,
- Removing surface water promptly by sweeping towards temporary drainage points. Plan that these are part of the design phase so that penetrations can be machined into the design. Locate drainage points in areas where the water will likely pond or can be easily bunded,
- Keeping materials off the surface and away from standing water,
- Reducing exposure time to the elements, timely sequencing and installing in the drier months can limit the risk,
- Sequencing façade installation with timber installation,
- Applying and maintaining end grain sealers,
- Applying and maintaining joint tapes and temporary vapour permeable membranes,
- Installing temporary flashings/membranes around column junctions and floor penetrations,
- Installing temporary rain covers over top of walls and columns to reduce water uptake, and
- Installing temporary impact protection in high traffic areas.

Drying of Timber and Finishes

It is recommended that drying of timber products should be conducted in a slow and controlled manner. Accelerated drying using external sources such as heaters can rapidly draw moisture from the timber and lead to surface cracking and checking which can affect the structural integrity of the timber elements.

Before applying any finishes (including plasterboard), the timber should be left to freely acclimatise and allow the moisture content to return to an equilibrium moisture level of between 10%-15%. Consult with the manufacturer of selected finishes for any further requirements.

If weathering occurs, it may be necessary to sand horizontal surfaces such as floors to remove surface cupping prior to the application of floor finishes.

For further reading, refer to Forestry Wood Products of Australia - ‘Woodsolutions Design Guide 53’.

NeXTimber® Temporary Protection

Temporary End Grain Sealer

There is the option for NeXTimber® products to come with a factory applied temporary sealer, specifically formulated to be applied to the end grain for the purpose of protecting against moisture absorption through the end grain encountered through normal weather exposure during construction.

Where agreed between the customer and NeXTimber® this can be applied to end grains, penetrations, and rebates. Note that temporary end grain sealers do not eliminate the uptake of moisture through the end grain but can reduce the amount of moisture absorbed through the end grains.

Note that this coating is only temporary, provides limited protection and should be used in conjunction with an overall site moisture management strategy.

Temporary Moisture Membranes

There is the option, if agreed between the customer and NeXTimber®, for NeXTimber® CLT to be supplied with a factory applied self-adhesive vapour permeable moisture membrane to the surface of the panels for the purpose of reducing moisture absorption encountered through normal weather exposure encountered during construction

For full effectiveness, additional tapes, flashings and termination details will need to be site applied by the customer to joints and interfaces in accordance with the membrane supplier's details and recommendations as well as intermittent construction maintenance (i.e through wear and tear, foot traffic etc).

NeXTimber® applied temporary coatings and membranes should be selected in conjunction with a well developed and implemented moisture management plan.

Please note that any vapour permeable membranes applied by NeXTimber® are only provided for the purpose of temporary protection, irrespective of whether the membrane applied is technically capable of ongoing protection.

This includes that the NeXTimber® processes for applying membranes have been considered only from a temporary protection perspective (i.e. during the transportation and construction phase) and under no circumstances are such membranes intended to provide ongoing protection beyond the transportation and construction phase.

In the event that the customer requires membranes which form part of the building envelope, it is the Customer's risk and responsibility to ensure that the membrane is fit for purpose and has been applied as required.

Pre-Assembly Checks

The following highlights some key items for the customer and its design team to check through the various stages of the documentation and assembly phases. The list is non-exhaustive and there may be additional items not listed.

Construction and Delivery Sequencing

Desired installation sequencing must be advised by the customer prior to the commencement of digital manufacturing modelling and panelisation. There may be instances where the installation sequence is dictated by;

- panel-to-panel connection details,
- panel set downs,
- finished floor levels and steps,
- panel geometry, and
- interactions with other building materials such as steelwork.

In these cases, the installation sequence will be adjusted during the manufacturing documentation phase to accommodate the sequence dictated by the nominated structural details.

During the Manufacturing Documentation Process

- Ensure the proposed installation sequence and site access have been coordinated with the delivery and logistics plan.
- If timber elements are interfacing with the existing structure, ensure that sufficient tolerances have been communicated to and that the dimensions have been validated on site including the verticality of the structure and any protrusions that may obstruct the installation.
- Ensure coordination with steelwork has been completed, including adequate tolerances, shelf angles have adequate bearing to accept screw fixings and that there is adequate access to install screws.
- Ensure the proposed temporary anchor/propping point locations have been coordinated with exposed timber finishes.
- Ensure the installation sequence and methodology of timber elements and steelwork have been considered and can occur unencumbered, particularly where elements intersect or sit within the same plane or within steelwork flanges.

Product Arrival Checklist

Upon receipt check;

- timber plugs have been provided for repairing sling holes where required.
- quantity and specification of supplied screws, components and connections are consistent with the take-off provided by NeXTimber® and are consistent with the engineer's structural specification.
- compounds and materials (i.e. fire caulking, acoustic strips) required between joint are onsite.

Site Assembly

Project Documentation and Connection Details

All timber connection details must be specified by the customer's structural engineer. This includes timber-to-timber connections, timber-to-steel connections, and timber-to-concrete/foundation.

Should there be any discrepancy, ambiguity or conflicting information between the structural engineering drawings and specifications and NeXTimber® documentation, then the customer should seek advice from its structural engineer and advise NeXTimber® accordingly.

To ensure that the products function as intended, refer to the fixings manufacturer for installation requirements, particularly reference for minimum edge distances and spacing, torque settings and any predrilling requirements.

During Installation

NeXTimber® products are highly accurate, manufactured building materials. Pay attention to details that contain tolerances and/or gaps. Ensure that any tolerances, gaps and clearances between panels and structural members have been accounted for in the setting out of panels as incorrect gapping in joints may lead to inconsistencies and unnecessary site rework.

- Continually check and confirm running dimensions against architectural gridlines and set out points.
- Install fixings as per the structural engineers' specifications as nominated on their structural documentation and per the requirements of the fixing supplier (i.e. drill and torque settings, minimum edge distance). Note that fixings supplied may come in varying lengths and sizes and ensure that the correct fixings are used in the correct locations. Seek clarity from the structural engineer where required.
- Ensure that approved caulking compounds or materials specified within joints (i.e. typically for fire and acoustic purposes) are installed during the installation process. Consult with the building certifier about what evidence/inspections are required for approval of connections and joints.

Temporary Works

The customer's temporary works engineer is typically responsible for maintaining stability of NeXTimber® elements and the building system during the erection and assembly, including but not limited to approval of panels for lifting (including the lifting system), placement and removal of temporary props and bracing and review of structural loading through the construction phase.

Lifting

The customers temporary works engineer should also review the panels (size, weight and geometry including openings and cutouts), proposed lifting system and lifter placement and ensure (at a minimum) that the lifters are equally placed around the centre of mass and that they are suitable for safe lifting and handling.

When reviewing the lifting points, ensure that the panels are balanced and that load is equally distributed to all lifting points.

To maintain panel stability and rigidity, parts of openings may be retained (e.g. strong backs in the form of part door sills and stiffening ribs) for stability during transportation and installation. These strong backs are to be removed by the customer with a panel saw once the panels are in their final position. The customer is responsible for the disposal of these items.

Temporary Propping and Bracing

Fixings and connections supplied with NeXTimber® products are for the end use condition only and are supplied in accordance with the customer's Design Documents and the Final Project Data.

Additional fixings and connections for the temporary condition, including the supply of temporary lifting devices, can be supplied and incorporated by NeXTimber® at the time of developing the manufacturing model if requested by the customer advising the relevant specification and quantity, and if practicable and agreed to by NeXTimber®.

Lifting and Manual Handling

Unless otherwise agreed with the customer, panels required in the final assembly of the project (ie. not offcut panels included within the delivery) will be supplied with nylon slings.

Timber plugs will be supplied with the NeXTimber® CLT panels to repair plug holes and will need to be fitted by the customer once the panel is in its final position. Should the penetration be within an element requiring a Fire Resistance Level (FRL), additional fire rated material needs to be installed by the customer inside the hole.

Refer to the fire report for details. When lifting with fabric slings, there may be some superficial marring around the lifting point which may require some remediation on site if the panel is to be left exposed.

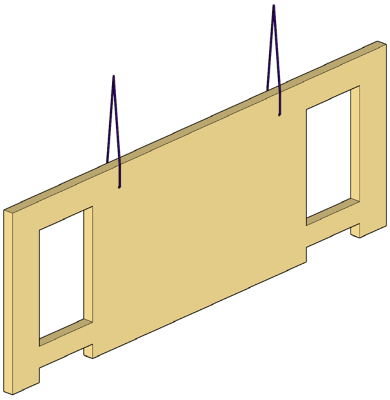

Figure 7: Nylon / fabric slings

Figure 8: Screw and clutch type arrangements (source: Rothoblaas)

Figure 9: Proprietary brackets and plates (source: Rothoblaas)

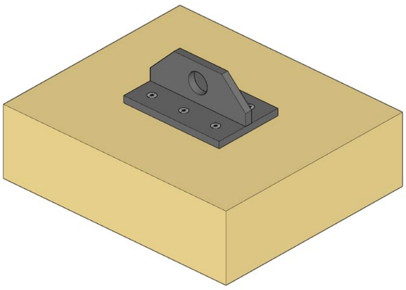

Figure 10: Customer designed/sourced lifting plates.

These may need to be procured directly by the customer if they are not a standard offer by NeXTimber® and selected lifting methods should be advised prior to the commencement of digital manufacturing modelling and documentation phase so that the correct machining details can be incorporated into the manufacturing documentation should they be required.

Panel-to-Panel Tolerances and Setting Out

NeXTimber® components are manufactured to tight tolerances. Excessive gaps in panel joints during installation can result in irregularities and unwarranted on-site adjustments.

Certain panel-to-panel connections may involve specific tolerances. While arranging the components, it is advisable for the customer and its design team to verify the specified tolerances nominated in the joints from the provided shop drawings and regularly check the set-out dimensions to ensure the assembly aligns with the provided documentation.

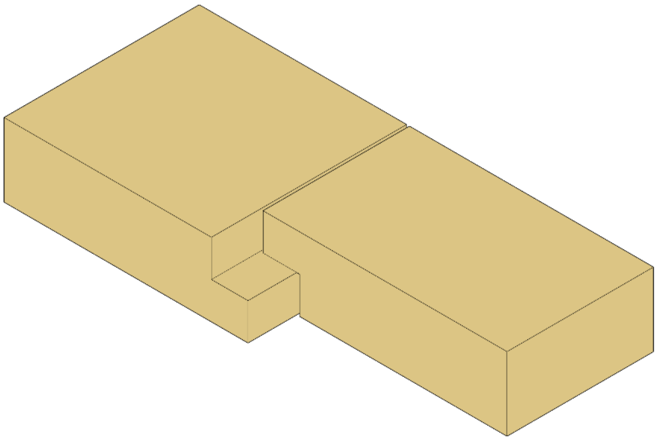

Figure 11: Example detail with tolerance.

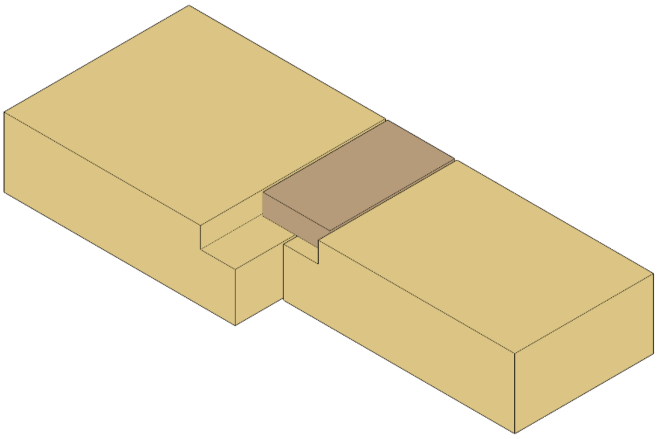

Figure 12: Example detail with tolerance.

Note: Fire rated caulking compounds, or acoustic treatments not shown in detail for clarity.

Shimming and Levelling

Shimming and leveling against concrete slabs may be necessary depending on the accuracy of the slab/substrate.

Any shimming should be done using non-metallic shims to prevent leaching or staining due to the development of surface rust. When installing, ensure that constant bearing is achieved under the panels by ram packing bearing joints with mortar mix. Avoid using wet mix mortar as the timber panels may absorb moisture from the mix.

Fire and Acoustic Requirements

Where the system requires an FRL, joints require a fire-resistant caulking which needs to be placed by the customer at the time of assembly. Refer to the *NeXTimber® Fire Performance Guide and RIR (Regulatory Information Report)* for caulking details.

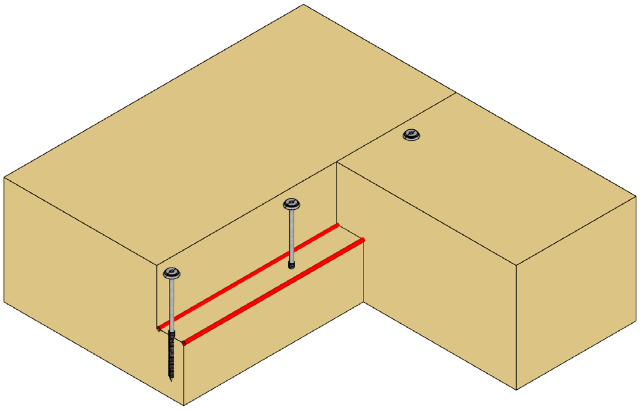

Figure 13: Typical lap connection detail.

Where acoustic strips or other barriers have been specified within the joints these need to be placed by the contractor at the time of assembly. Refer to the architects' drawings for details and specifications for any additional details.

Modifications to NeXTimber® Products

NeXTimber® products are structural products, and any modifications or rectifications must not be undertaken without prior approval from the customer's structural engineer. This includes chasing, cutting, or remediating.

Site Inspections and Installation Sign-off

NeXTimber® is not responsible for and will not undertake structural site inspections or provide sign-off on installation or site management.

Assembly Site Tools

Several tools are available on the market to assist in installation, some examples of which include the following.

Figure 14: Pry Bar (source: Rothoblaas)

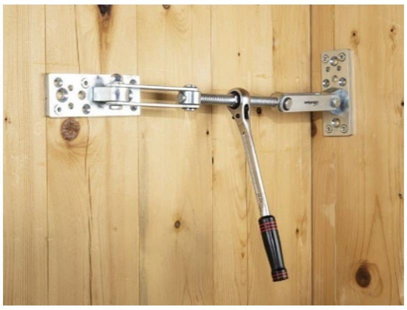

Figure 15: 'Geko Panel Puller' (source: Rothoblaas)

Figure 16: 'Skorpio Beam Puller' (source: Rothoblaas)

Figure 17: Giraffe Panel props (source: Rothoblaas)

Figure 18: Cordless Drills / Drivers (source: Rothoblaas)

It starts with a seed from nature, with soil, water and sunshine added.

It's what a better tomorrow will be built on, built out of, built with.

It's what the craftsmen, the draftsmen, the architect, the engineer – all those who shape tomorrow – should lean on.

If it can be renewable and store carbon, it should, and it can.

And if it can all be done quicker, quieter, smarter, and safer it should.

And it can.

You see, if we treat our buildings as living breathing things, then the living breathing things that inhabit them, will have a future to look forward to.

Everything we do today matters, for tomorrow will be where we all reside.

However you use NeXTimber, you won't just be building a building.

You'll be creating a better future.

NeXTimber®
by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

nextimber.com.au

Version 1.0_072024

CLT and GLT illustrations used throughout are artist impressions